

RRB ALP CBT 2 (Part A) Master Question Bank

Section 1: Basic Science and Engineering

Engineering Drawing and Projections

Q1. Which type of line is used to indicate hidden edges and surfaces in an engineering drawing?

Ans: Dashed narrow line (or hidden line).

Q2. In a third-angle projection, where is the top view placed relative to the front view?

Ans: Directly above the front view.

Q3. What is the standard size of an A4 drawing sheet in millimeters?

Ans: 210 mm x 297 mm.

Q4. Which scale is used when the drawing dimensions are exactly the same as the actual dimensions of the object?

Ans: Full size scale (1:1).

Q5. A curve generated by a point fixed to a circle as the circle rolls along a straight line without slipping is called a _____.

Ans: Trochoid (or Cycloid if on the circumference).

Units and Measurements

Q6. What is the SI unit of luminous intensity?

Ans: Candela (cd).

Q7. Convert -40 degrees Celsius into Fahrenheit.

Ans: -40 degrees Fahrenheit. (The temperatures are equal at this point).

Q8. One horsepower (HP) in the metric system is approximately equal to how many watts?

Ans: 735.5 Watts (Note: British system is 746 W).

Q9. Which instrument is used to measure the specific gravity of liquids?

Ans: Hydrometer.

Mass, Weight and Density

Q10. If the mass of an object is 60 kg on Earth, what will be its mass on the Moon?

Ans: 60 kg. (Mass remains constant everywhere; only weight changes).

Q11. What is the relationship between density, mass, and volume?

Ans: $\text{Density} = \text{Mass} / \text{Volume}$.

Q12. What is the density of pure water at 4 degrees Celsius?

Ans: 1000 kg/m^3 (or 1 g/cm^3).

Speed, Velocity and Work, Power, Energy

Q13. What does the area under a velocity-time (v-t) graph represent?

Ans: Displacement.

Q14. A body of mass 10 kg is dropped from a height of 20 m. Find its kinetic energy just before hitting the ground. (Take $g = 10 \text{ m/s}^2$)

Ans: 2000 Joules. (PE lost = KE gained = $m * g * h = 10 * 10 * 20$).

Q15. If a force of 50 N moves a body by 5 m in the direction of the force, what is the work done?

Ans: 250 J (Work = Force * Displacement).

Q16. What is the commercial unit of electrical energy, and how many Joules does it equal?

Ans: Kilowatt-hour (kWh); $1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$.

Heat and Temperature

Q17. By which mode is heat transferred through a solid metal rod?

Ans: Conduction.

Q18. What is the amount of heat required to raise the temperature of 1 g of water by 1 degree Celsius called?

Ans: 1 Calorie.

Q19. Why does a person feel colder after coming out of a swimming pool on a windy day?

Ans: Due to the cooling effect caused by the rapid evaporation of water from the skin.

Levers and Simple Machines

Q20. What is the mechanical advantage (MA) formula for a simple lever?

Ans: $MA = \text{Effort Arm Length} / \text{Load Arm Length} = \text{Load} / \text{Effort}$.

Q21. A pair of scissors or a crowbar belongs to which class of lever?

Ans: Class 1 Lever (Fulcrum is located between the load and the effort).

Q22. Which class of lever always has a mechanical advantage less than 1 ($MA < 1$)?

Ans: Class 3 Lever (e.g., human forearm, sugar tongs).

Basic Electricity

Q23. According to Kirchhoff's Current Law (KCL), what is the algebraic sum of currents meeting at a node?

Ans: Zero (It is based on the law of conservation of charge).

Q24. Three resistors of 6 ohms each are connected in parallel. What is their equivalent resistance?

Ans: 2 ohms. ($1/R_p = 1/6 + 1/6 + 1/6 = 3/6$, therefore $R_p = 6/3 = 2 \text{ ohms}$).

Q25. Which device is used to measure electrical current in a circuit, and how is it connected?

Ans: Ammeter; connected in series.

Occupational Safety, Health and Environment

Q26. Which class of fire involves electrical equipment, and which extinguisher should be used?

Ans: Class C (or Class E in some standards); Carbon Dioxide (CO₂) or Halon/Dry Powder extinguishers should be used.

Q27. What does the "Kyoto Protocol" international treaty aim to reduce?

Ans: Greenhouse gas emissions to combat global warming.

Q28. What type of Personal Protective Equipment (PPE) protects workers against penetrating objects like nails/spikes?

Ans: Safety shoes with steel toes/soles (Foot protection).

Section 2: Mathematics

Q29. Find the value of x: Square root of $(5x - 4) = 6$.

Ans: $x = 8$. (Squaring both sides: $5x - 4 = 36$, which means $5x = 40$).

Q30. If the ratio of two numbers is 3:4 and their HCF is 4, what is their LCM?

Ans: 48. (LCM = $3 * 4 * \text{HCF} = 3 * 4 * 4$).

Q31. Simplify using BODMAS: $25 - [20 - \{10 - (7 - 5)\}]$.

Ans: 13.

Q32. A train 150 m long passes a telegraph post in 9 seconds. Find the speed of the train in km/h.

Ans: 60 km/h. (Speed = $150/9 \text{ m/s} = 150/9 * 18/5 = 60 \text{ km/h}$).

Q33. A shopkeeper buys an article for 800 rupees and sells it for 1000 rupees. Find his profit percentage.

Ans: 25%. (Profit % = $200 / 800 * 100$).

Q34. If A:B = 2:3 and B:C = 4:5, find A:B:C.

Ans: 8:12:15.

Q35. Find the compound interest on 10,000 rupees for 2 years at 10% per annum, compounded annually.

Ans: 2,100 rupees.

Q36. Evaluate: $\sin^2(30) + \cos^2(30)$.

Ans: 1. (Standard trigonometric identity $\sin^2(\theta) + \cos^2(\theta) = 1$).

Q37. The mean of 5 numbers is 20. If one number is excluded, the mean becomes 18. Find the excluded number.

Ans: 28. (Original total = 100, New total = 72, so $100 - 72 = 28$).

Q38. Two pipes A and B can fill a tank in 12 hours and 15 hours respectively. If both open together, how long will they take?

Ans: 6 hours 40 minutes (or 6 and $\frac{2}{3}$ hours). ($\frac{1}{12} + \frac{1}{15} = \frac{9}{60} = \frac{3}{20}$).

Q39. Find the total surface area of a solid hemisphere of radius 7 cm.

Ans: 462 cm². (TSA = $3 * \pi * r^2 = 3 * \frac{22}{7} * 7 * 7$).

Q40. Solve for x: $x^2 - 5x + 6 = 0$.

Ans: $x = 2$ and $x = 3$.

Section 3: General Intelligence and Reasoning

Q41. Complete the series: 4, 9, 16, 25, 36, ?

Ans: 49. (Perfect squares sequence: $2^2, 3^2, 4^2, 5^2, 6^2, 7^2$).

Q42. In a certain code, 'ROSE' is written as 'TQUG'. How is 'BISCUIT' written in that code?

Ans: DKUEWKV. (Each letter is shifted forward by +2 positions).

Q43. Pointing to a man, a woman said, "His mother is the only daughter of my mother." How is the woman related to the man?

Ans: Mother.

Q44. If 'A' means '+', 'B' means '-', 'C' means '*', and 'D' means '/', then what is the value of 18 C 14 A 6 D 3?

Ans: 254. ($18 * 14 + 6 / 3 = 252 + 2 = 254$).

Q45. A man walks 8 km South, then turns West and walks 6 km. How far is he from his starting point?

Ans: 10 km. (Using Pythagoras theorem: Square root of ($8^2 + 6^2$) = Square root of 100).

Q46. Choose the odd one out: Earth, Mars, Moon, Venus.

Ans: Moon. (Moon is a natural satellite, whereas others are planets).

Q47. If 1st January 2024 was a Monday, what day of the week was 1st January 2025?

Ans: Wednesday. (2024 was a leap year, so it adds 2 odd days).

Q48. Statements: All pens are books. All books are markers.

Conclusion: I. All pens are markers. II. Some markers are pens. Which follows?

Ans: Both conclusions I and II follow.

Q49. Arrange in a logical order: 1. Engine, 2. Ticket, 3. Platform, 4. Train, 5. Journey.

Ans: 2, 3, 4, 1, 5 (Ticket -> Platform -> Train -> Engine -> Journey).

Q50. Clock Question: What is the angle between the hour hand and minute hand at 4:20?

Ans: 10 degrees. (Using formula: Angle = $|30H - 5.5M| = |30*(4) - 5.5*(20)| = |120 - 110| = 10$ degrees).

Engineering Drawing & Projections

Q51. What is the standard angle of inclination used for lettering in engineering drawings?

Ans: 75 degrees.

Q52. Which type of projection uses a single vanishing point?

Ans: One-point perspective projection.

Q53. If a line is parallel to the Horizontal Plane (HP) and perpendicular to the Vertical Plane (VP), what will its top view look like?

Ans: A straight line showing its true length.

Q54. Which instrument is used to draw smooth curves that cannot be drawn with a standard compass?

Ans: French curves.

Units & Measurements

Q55. What is the SI unit of electric charge?

Ans: Coulomb (C).

Q56. One astronomical unit (AU) represents the average distance between which two celestial bodies?

Ans: The Earth and the Sun.

Q57. Express 0 Kelvin in terms of the Celsius scale.

Ans: -273.15 degrees Celsius.

Mechanics & Properties of Matter

Q58. What is the term used for the ratio of shear stress to shear strain within the elastic limit?

Ans: Modulus of Rigidity (or Shear Modulus).

Q59. According to Hooke's Law, stress is directly proportional to strain up to which point?

Ans: Proportional limit (or Elastic limit).

Q60. What is the mechanical advantage of a fixed pulley?

Ans: 1. (It only changes the direction of the force, not its magnitude).

Q61. A machine has an efficiency of 80%. If the input energy is 500 Joules, what is the useful output work?

Ans: 400 Joules. (Output = Efficiency * Input = 0.80 * 500).

Heat & Basic Electricity

Q62. What is the value of the specific heat capacity of water in SI units?

Ans: 4200 J/kg-K (or 4.2 J/g-degree Celsius).

Q63. Which law states that the current flowing through a conductor is directly proportional to the potential difference across its ends, provided temperature remains constant?

Ans: Ohm's Law.

Q64. If a wire of resistance R is stretched to double its original length while keeping its volume constant, what will its new resistance be?

Ans: $4R$. (Resistance is proportional to the square of the length when stretched).

Section 2: Mathematics (Additional)

Q65. Find the value of x if 25% of x is equal to 40.

Ans: $x = 160$. ($0.25 * x = 40$, so $x = 40 / 0.25$).

Q66. The ratio of the ages of A and B is 4:5. If the sum of their ages is 36 years, find the age of A.

Ans: 16 years. ($4x + 5x = 36$, meaning $9x = 36$, so $x = 4$. A's age = $4 * 4 = 16$).

Q67. A person crosses a 600-meter long street in 5 minutes. What is his speed in km/h?

Ans: 7.2 km/h. (Speed = $600\text{m} / 300\text{s} = 2 \text{ m/s}$. Convert to km/h: $2 * 18/5 = 7.2 \text{ km/h}$).

Q68. If the radius of a circle is increased by 10%, by what percentage will its area increase?

Ans: 21%. (Using net percentage formula: $10 + 10 + (10*10)/100 = 21\%$).

Q69. Find the value of $(\tan 45^\circ + \cot 45^\circ)$.

Ans: 2. (Since $\tan 45^\circ = 1$ and $\cot 45^\circ = 1$).

Q70. If $\log(x) + \log(2) = \log(8)$, find the value of x .

Ans: $x = 4$. ($\log(2x) = \log(8)$, therefore $2x = 8$, so $x = 4$).

Section 3: General Intelligence & Reasoning (Additional)

Q71. Complete the number series: 2, 6, 12, 20, 30, ?

Ans: 42. (The differences between consecutive terms are +4, +6, +8, +10, so the next difference is +12).

Q72. If in a certain code language, 'BLUES' is written as 'HVAKY', how will 'GREEN' be written?

Ans: MXKKT. (Each letter is shifted forward by +6 positions).

Q73. A man faces North. He turns 45 degrees clockwise, then 90 degrees counter-clockwise. Which direction is he facing now?

Ans: North-West.

Q74. Introducing a boy, a girl says, "He is the son of the only son of my maternal grandfather." How is the boy related to the girl?

Ans: Cousin (or Brother, depending on options, but structurally Cousin via maternal uncle).

Q75. Statement: Should physical education be made compulsory in schools?

Arguments: I. Yes, it helps in the overall physical and mental development of students. II. No, it takes away valuable academic study time. Which argument is strong?

Ans: Only argument I is strong.

Engineering Drawing & Instruments

Q76. Which type of scale is used to read three consecutive dimensions, such as meters, decimeters, and centimeters?

Ans: Diagonal scale.

Q77. What is the standard angle of the lines used in hatching (sectioning) an object in engineering drawing?

Ans: 45 degrees.

Q78. In an isometric drawing, what is the angle between the three principal axes?

Ans: 120 degrees.

Q79. Which drawing instrument consists of a stock and a blade, used specifically to draw horizontal parallel lines?

Ans: T-Square.

Mechanics & Properties of Matter

Q80. What is the SI unit of coefficient of friction?

Ans: It has no unit (it is a dimensionless quantity).

Q81. If a body is floating in a fluid, its buoyant force must be equal to what?

Ans: The weight of the fluid displaced by the body (Archimedes' Principle).

Q82. What is the maximum value of static friction called?

Ans: Limiting friction.

Q83. A machine requires an effort of 20 N to lift a load of 80 N. What is the mechanical advantage (MA) of this machine?

Ans: 4. ($MA = \text{Load} / \text{Effort} = 80 / 20$).

Heat, Temperature & Electricity

Q84. What is the phenomenon where heat is transferred through the actual movement of fluid molecules from hotter to colder regions?

Ans: Convection.

Q85. What is the resistance of an ideal voltmeter and an ideal ammeter?

Ans: Ideal Voltmeter = Infinite resistance; Ideal Ammeter = Zero resistance.

Q86. If two bulbs of 40 Watts and 60 Watts are connected in series, which bulb will glow brighter?

Ans: The 40 Watt bulb. (In series, the bulb with higher resistance develops more power and glows brighter).

Q87. What is the unit used to measure the capacity of a battery (like those used in locomotives or cars)?

Ans: Ampere-hour (Ah).

Section 2: Mathematics (Expected)

Q88. Find the single equivalent discount for two successive discounts of 20% and 10%.

Ans: 28%. (Using formula: Net Discount = $x + y - (x * y) / 100 = 20 + 10 - 2 = 28\%$).

Q89. If a sum of money doubles itself in 5 years at simple interest, what is the rate of interest per annum?

Ans: 20%. (Interest = Principal, so $R = (100 * I) / (P * T) = 100 / 5 = 20\%$).

Q90. A man covers a certain distance at 40 km/h and returns at 60 km/h. Find his average speed for the whole journey.

Ans: 48 km/h. (Average Speed = $2 * v_1 * v_2 / (v_1 + v_2) = 2 * 40 * 60 / 100 = 48$).

Q91. If the length of a rectangle is increased by 20% and its breadth is decreased by 10%, what is the net change in its area?

Ans: 8% increase. (Using net percentage: $20 - 10 - (20 * 10) / 100 = 10 - 2 = 8\%$).

Q92. Evaluate: $(1 - \sin^2 \theta) * \sec^2 \theta$.

Ans: 1. (Since $1 - \sin^2 \theta = \cos^2 \theta$, and $\cos^2 \theta * \sec^2 \theta = 1$).

Q93. The HCF and LCM of two numbers are 12 and 72 respectively. If one of the numbers is 24, find the other number.

Ans: 36. (First Number * Second Number = HCF * LCM, so $24 * x = 12 * 72 \rightarrow x = 36$).

Section 3: General Intelligence & Reasoning (Expected)

Q94. Complete the letter series: Y, W, U, S, Q, ?

Ans: O. (The letters are moving backward by skipping one letter: -2 positions).

Q95. In a row of 35 students, Ramesh is 18th from the left end. What is his position from the right end?

Ans: 18th. (Total = Left + Right - 1 $\rightarrow 35 = 18 + \text{Right} - 1 \rightarrow \text{Right} = 18$).

Q96. If it is 3:00 o'clock on a standard wall clock, what is the angle between the hour hand and the minute hand?

Ans: 90 degrees.

Q97. Choose the odd number out: 27, 64, 125, 144, 216.

Ans: 144. (144 is a perfect square [12^2], while all the others are perfect cubes [3^3 , 4^3 , 5^3 , 6^3]).

Q98. Statements: Some doctors are teachers. All teachers are counselors.

Conclusion: I. Some doctors are counselors. II. Some counselors are teachers. Which follows?

Ans: Both conclusions I and II follow.

Q99. Looking at a portrait, Advait said, "Her mother is the wife of my father's only son." Whose portrait is Advait looking at?

Ans: His daughter's portrait.

Q100. If total 11 letters of the English alphabet are written in capital letters, how many of them do not change their shape when viewed in a vertical mirror?

Ans: 11 letters. (The letters are: A, H, I, M, O, T, U, V, W, X, Y).